

Qns	Answer	Marks
7(a)(i)	work done per unit mass in bringing a small test mass from infinity to that point	B1 B1
7(a)(ii)	potential is zero at infinity Gravitational force is attractive work is done by (two) masses in moving them closer together work is done on (two) masses in moving them apart test mass getting closer from infinity loses potential energy	Any 2
7(b)	Loss in kinetic energy = Gain in Gravitational Potential Energy Initial KE – final KE = final GPE – initial GPE $\frac{1}{2}mv^2 - 0 = 0 - m\phi$ $v = \sqrt{-2\phi}$	B1 B1 B1 A0
7(c)	$\phi = -GM / r$ $= - (6.67 \times 10^{-11} \times 7.3 \times 10^{22}) / (1.7 \times 10^6)$ $= - 2.9 \times 10^6 \text{ J kg}^{-1}$	C1 A1
7(d)	speed = $\sqrt{2 \times 2.9 \times 10^6}$ $= 2400 \text{ m s}^{-1}$	A1
7(e)	$\frac{1}{2}m\langle c^2 \rangle = (3/2)kT$ $\frac{1}{2} 3.34 \times 10^{-27} \times \langle c^2 \rangle = (3/2) (1.38 \times 10^{-23})(400)$ $c_{r.m.s.} = 2200 \text{ m s}^{-1}$	C1 C1 A1
7(f)	r.m.s. speed is an average so many molecules have speeds greater than the escape speed or there is a distribution of molecular speeds (around the r.m.s. value) so many molecules have speeds greater than the escape speed	B1
7(g)(i)	Loss in GPE = Gain in KE $\left(-\frac{GMm}{r_x}\right) - \left(-\frac{GMm}{r_y}\right) = \frac{1}{2}mv_y^2 - \frac{1}{2}mv_x^2$ $(6.67 \times 10^{-11})(1.99 \times 10^{30}) \left[\left(-\frac{1}{8.44 \times 10^{11}}\right) + \left(\frac{1}{6.38 \times 10^{10}}\right) \right] = \frac{1}{2}v_y^2 - \frac{1}{2}(34.1 \times 10^3)^2$ $v_y = 70.8 \text{ km s}^{-1}$	B1 C1 C1 A1
7(g)(ii)	both Gravitational Potential Energy and Kinetic Energy equations include m , so path is unchanged OR It is independent on m , so path is unchanged.	B1 (B1)

Qns	Answer	Marks
7(g)(iii)	Path higher than the original one AND Closest distance of approach is larger And if goes back, the bottom path is lower than the original one.	B1